

SLOWLAB: Benchmarking Scientific Agents When Experiments Are Slow, Noisy, Costly, and Irreversible

Anonymous Authors

Abstract

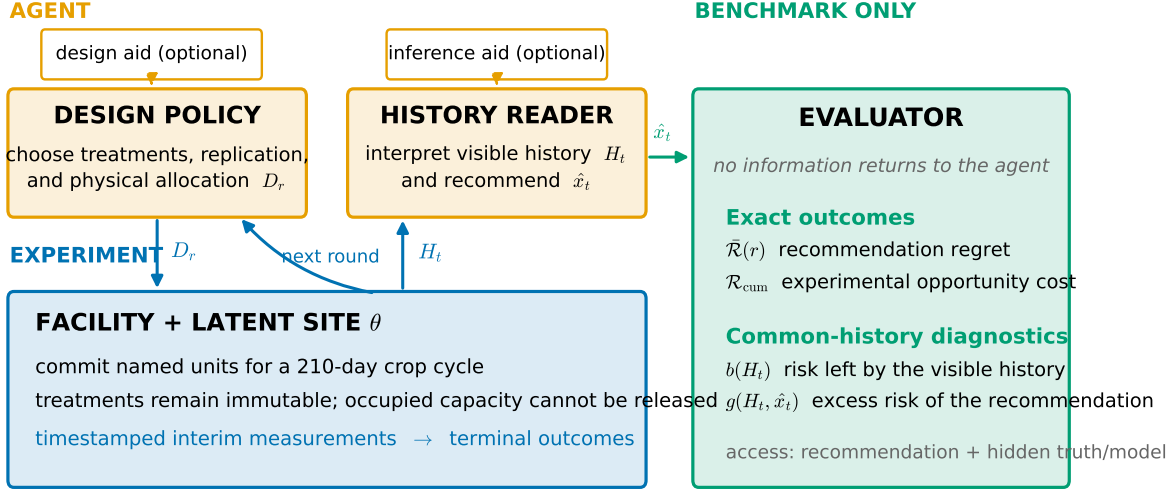
Scientific-agent benchmarks rarely test whether an agent can allocate a fixed set of physical units, use evidence that arrives during long-running trials, and commit to a final operating recommendation whose experimental cost is paid in the objective itself. We introduce SLOWLAB, an executable controlled-environment horticulture benchmark with split-plot controls, capped parallel capacity, correlated noise, irreversible 210-day crop cycles, time-appropriate interim observations, and separate design and recommendation actions. We evaluate final and cumulative regret and use a common-history Bayes-risk decomposition to diagnose what an executed design supports and how much excess risk remains in the agent’s recommendation; independent posterior samples avoid in-sample action selection, and the calibrated one-factor case has response-curve RMSE 0.00445. Across 928 episodes under a frozen protocol, neither design nor inference aids significantly changed final regret for Luna on 56 paired sites, and an inference aid did not improve Sol on 24 further sites; a fresh 384-episode extension found the same null result at twice the observation noise, while a prespecified secondary analysis found that the design aid increased regret at half noise ($\Delta = 0.0113$, 95% CI [0.0053, 0.0184], adjusted $p = 0.0096$).

1 Introduction

Language-model agents are increasingly proposed as operators of scientific experimentation: selecting what to run, interpreting what comes back, and deciding what to run next [Boiko et al., 2023, M. Bran et al., 2024, Szymanski et al., 2023]. The claim attached to this work is that such agents will accelerate discovery in domains where experiments are the bottleneck. Whether that claim holds is an empirical question, and answering it requires evaluating agents under the conditions that make experimentation a bottleneck in the first place.

Existing evaluations cover important parts of this loop. BoxingGym limits an agent to ten sequential queries and scores information gain and prediction; Olympus and Summit provide executable optimization surfaces derived from chemistry; Minerva studies noisy parallel batches under laboratory constraints; and budgeted and delayed-feedback Bayesian optimization formalize cost or latency [Gandhi et al., 2025, Häse et al., 2021, Felton et al., 2021, Sin et al., 2025, Astudillo et al., 2021, Verma et al., 2022]. What remains difficult to measure is the combined decision: allocating treatments to a tiered physical facility, interpreting the resulting history, and issuing an operating recommendation while the experiments themselves incur opportunity cost. A commercial tomato crop makes this combination concrete. It occupies a greenhouse for seven months, block-level climate controls restrict which treatments can run together, and a committed treatment cannot be revised after later evidence arrives.

SLOWLAB. We introduce an executable benchmark for experimental design under four conditions that hold at once. Feedback is **slow**: a cycle is 210 days, and the number of trials that run concurrently is capped — unevenly, because some factors are set for a whole block of units and others per unit, so replication and coverage compete for the same physical resource. Observations are **noisy** enough that one unit does not settle a question. Experimentation is **costly** in the objective’s own units, since testing a management regime means growing a crop under it and selling whatever results. And a submitted design is **irreversible**: no branch, no checkpoint, no revert. Each condition alone is familiar. Together they withdraw common forms of recourse, because an agent can neither buy precision and coverage from the same units nor find out which it needed before the season ends.



A closed-loop agent may implement design and reading in one model; fixed-history replay separates the roles experimentally.

Figure 1: **Roles and information flow.** The design policy commits treatments and physical units; the history reader maps only the time-stamped visible history to a recommendation. One closed-loop agent may implement both roles. The evaluator belongs to the benchmark: it receives the recommendation and hidden truth to compute exact outcomes, and uses the same visible history for diagnostic risks. It never supplies hidden information to the agent. Interim observations can update the reader and the next design, but cannot alter committed treatments or release occupied capacity.

We validate this regime in controlled-environment horticulture. A published mechanistic crop model [Jones et al., 1999] supplies a traceable physiological basis, and an explicit economic model puts recommendations and experimental opportunity cost in one currency, net profit per square metre per day (§3.1). The published model does not make benchmark difficulty author-independent: factor ranges, facility size, task budgets and observation channels remain design choices and are reported as such. We distinguish three roles throughout. A *design policy* chooses which factor settings to test, on how many units, and which physical units those are. Lamps and heating serve a whole chamber while planting density and the pruning target are set per hydroponic loop, so an allocation that varies temperature between two loops of one chamber describes an experiment that cannot be performed. A *history reader* turns the time-stamped evidence into a recommendation. In a closed-loop LLM these two roles live in the same model, but we separate them experimentally. The benchmark-side *evaluator* scores their outputs using hidden truth and never returns that information to the agent. Having committed, the agent may inspect measurements that are physically available as the crop develops, but it cannot observe future yield or profit, revise the submitted treatment, or release occupied capacity. After two or three cycles it must recommend. Figure 1 shows the campaign loop.

Overall, our work makes the following contributions:

1. We formulate agent evaluation as three coupled capabilities in a resource-limited batch campaign: allocating treatments to feasible physical units, using the evidence revealed by the executed campaign, and issuing a final recommendation. The greenhouse instantiation enforces split-plot controls, finite capacity, economic accounting and time-appropriate inspection during a committed crop cycle.
2. We define a common-history diagnostic. Bayes risk $b(H_t)$, recommendation risk and posterior excess $g(H_t, \hat{x}_t)$ condition on exactly the time-stamped history visible to the agent. Independent action-selection and evaluation samples prevent the evaluator from grading its own fitted action in sample.
3. We separate design from history reading by replaying multiple readers on fixed executed histories, then test design and inference aids in a frozen, site-paired protocol. On the tested Luna and Sol conditions the aids do not reliably reduce final regret; the result survives a fresh twice-noise Luna extension, while

the lower-noise secondary condition identifies a harmful design-aid effect.

2 SLOWLAB

2.1 Regime and problem formulation

SLOWLAB targets sequential experimental design under four conditions that hold jointly. **Slow**: what can be learnt per unit of calendar time is bounded by feedback latency *and* by a hard ceiling on how many trials run concurrently—a facility with a thousand chambers and a four-month cycle is not slow in the sense that binds. That ceiling is *tiered*, because some factors are set for a whole block of units and others per unit, so replication and coverage compete for the same physical resource. **Noisy**: a single unit does not settle a question, which is what turns replication, blocking and allocation into decisions rather than formalities. **Costly**: a trial consumes resources measured in the objective’s own units, so exploration is priced rather than free. **Irreversible**: a submitted design cannot be rolled back—no branch, no checkpoint, no revert—so the only remedy for a bad design is a new one in a later cycle, and the units it occupied are gone. Table 2 compares the published interfaces of related environments without treating unreported properties as absent. We instantiate the regime in controlled-environment agriculture (§3.1).

Problem formulation. An instance is a parameter vector $\theta \sim P_\Theta$, unknown to the agent, determining a response $f_\theta : \mathcal{X} \rightarrow \mathbb{R}$ on a factor space $\mathcal{X} \subseteq \mathbb{R}^d$ and measured in the objective’s own units. Over R rounds the agent commits a *design* $D_r \in \mathcal{D}$ — a finite set of treatments in $\mathcal{X}$ together with their assignment to the physical trial slots $\mathcal{U}_r$ — and states a recommendation $\hat{x}_r \in \mathcal{X}$. The admissible set $\mathcal{D}$ belongs to the facility rather than to the task; §4.1.3 gives it for our instantiation. Its one structural role here is that it is not all finite multisets: an agent cannot buy precision and coverage with the same slots.

Committing holds $\mathcal{U}_r$ for a cycle of length Δ , after which each slot returns one observation

$$y_u = f_\theta(x_u; \phi_u) + \varepsilon_u, \quad \varepsilon \sim \mathcal{N}(0, \Sigma(D_r)), \quad (1)$$

where x_u is the treatment assigned to slot u and ϕ_u an independent per-slot draw of the response model’s own parameters. Two features of (1) carry **noisy**. Drawing ϕ per slot rather than adding a scalar error makes the variance heteroscedastic in x and structured by the mechanism; and $\Sigma(D_r)$ is not diagonal, so *where* a treatment is placed changes what the round can resolve, not only which treatments were chosen. Appendix G gives Σ for our instantiation.

The other three conditions restrict the action. Writing $\mathcal{F}_r = \sigma(\{y_u : u \in \mathcal{U}_1 \cup \dots \cup \mathcal{U}_{r-1}\})$ for everything returned before round r , and $\mathcal{A}_r$ for the entire action set available at round r ,

$$D_r, \hat{x}_r \text{ are } \mathcal{F}_r\text{-measurable,} \quad \mathcal{A}_r = \{(D_r, \hat{x}_r)\}. \quad (2)$$

The first is **slow** stated as measurability rather than as sensor deprivation: information about a committed slot may well exist before its cycle ends, but no action is available on it, and Appendix J names the two actions that would separate the two readings. The second is **irreversible**: the action set contains commit and recommend and nothing else, from which it follows that no operation shrinks $\mathcal{F}_r$, releases a slot early, or refunds one already committed. **Costly** is carried not by a constraint but by the objective, which prices a trial in the units it is trying to maximise, so that a campaign has a cost beside the quality of its answer (§3.3).

The problem. Given $\mathcal{F}_r$ and nothing else, the agent chooses $D_1, \dots, D_R$ and finally commits to $\hat{x}_R$. It is judged on how far that recommendation falls short of the instance’s own optimum,

$$\mathcal{R} = f_\theta(x_\theta^*) - f_\theta(\hat{x}_R), \quad x_\theta^* \in \arg \max_{x \in \mathcal{X}} f_\theta(x), \quad (3)$$

and, separately, on what the campaign spent reaching it: every slot it committed is charged the same shortfall, at the treatment that slot was given, for a full cycle (5). Neither quantity is observable to the agent — θ , f_θ and x_θ^* are all withheld — and neither can be estimated from $\mathcal{F}_r$ without the design that generated it.

Three roles and their information boundary. We use role names rather than requiring three software modules. The *design policy* maps the visible history and remaining resources to D_r ; it determines what evidence can exist. The *history reader* maps that same visible history to beliefs and $\hat{x}_r$; it determines how acquired evidence is used. A closed-loop agent normally implements both. We separate them only for diagnosis: copying one accepted, completed history to several readers changes evidence use without changing the evidence, while holding the reader fixed across histories changes what the designs made observable.

The *evaluator* is benchmark-side. It receives submitted designs and recommendations and may access θ , f_θ , P_Θ and the forward model to compute exact outcomes or calibrated diagnostic quantities. None of those privileged objects enters the agent’s history, and evaluator output is not fed back during an episode. Thus a design aid can change D_r , an inference aid can change the reader’s interpretation and $\hat{x}_r$, and neither aid is an evaluator.

The two pull against each other, and the constraints are what stop the usual resolution. Reducing (3) calls for slots spent on treatments the agent expects to be poor, since those are the ones that discriminate; that is exactly what the second charge penalises. In an environment where a trial is a function call, the tension is nominal: try something, look, retract it. Here (2) removes retraction, (1) means one slot does not settle whether a treatment was poor, and $\mathcal{D}$ means the slots spent discriminating are slots not spent confirming. A campaign therefore occupies the facility for $R\Delta$ and consists of R ordered, unrepeatable commitments, each made before the information that would have justified it exists. §3.3 states both quantities precisely and what they are read against; Appendix B works one round through in full.

3 Tasks and evaluation

3.1 Controlled-environment horticulture

The current benchmark contains one validated domain. Its response model is the reduced state-variable TOMGRO model [Jones et al., 1999]; a site seed draws crop and economic parameters and fixes the corresponding response surface. Climate chambers are blocks and independently plumbed hydroponic loops are units. Temperature, photoperiod, lamp intensity and carbon-dioxide enrichment are chamber-level controls; planting density and the pruning target are loop-level controls. The objective is net profit per square metre per day. This construction makes the simulator traceable to a published crop model, while Appendix H records the additional choices made for parameter distributions, factor ranges, prices, observation noise and facility size.

3.2 Tasks

Four tasks vary the decision geometry while retaining the same crop-cycle duration, commitment rule and economic objective. SANITY is a one-factor interface and evaluator check. SCREEN allocates a sparse two-round budget across all six factors. OPTIMISE varies one chamber-level and one loop-level factor, so a feasible batch is a split-plot allocation rather than an unordered list of points. TRANSFER optimizes four resource factors and then asks for a new recommendation after electricity and gas prices rise by 2.2×, without allowing another experiment. Table 1 states the resource structure directly; we do not reuse the Version 1 task-difficulty ratios.

3.3 Evaluation

Following the information boundary of §2.1, evaluation has two layers. The *outcome evaluator* uses the hidden site to score what the campaign did: recommendation regret and the opportunity cost of its executed experiments. These exact simulator outcomes support the primary comparisons. The *common-history evaluator* is a calibrated benchmark-side diagnostic: it conditions on the same H_t visible to the reader, while using the known instance distribution and forward model to ask what that history could have supported. It never supplies a posterior, action or score to the agent. The layers are reported separately and never combined.

3.3.1 Realised regret

Both quantities of §2.1 are made precise here. They are regret against the same optimum in the same currency, because the experiment is paid for in the units of what it is trying to maximise. The first is (3),

Task	factors	rounds	units/round	unit-days	decision tested
Sanity	1	2	8	3,360	interface / calibration
Screen	6	2	12	5,040	scarce allocation
Optimise	2	3	12	7,560	split-plot iteration
Transfer	4	3	24	15,120	post-shock recommendation

Table 1: Version 2 tasks. Unit-days are the physical capacity committed by one campaign; factor ranges and control levels are given in Appendix C.

elicited after *every* round rather than only at R , and averaged over instances:

$$\bar{\mathcal{R}}(r) = \mathbb{E}_{\theta \sim P_{\Theta}} \left[f_{\theta}(x_{\hat{\theta}}^*) - f_{\theta}(\hat{x}_r) \right], \quad x_{\hat{\theta}}^* \in \arg \max_{x \in \mathcal{X}} f_{\theta}(x). \quad (4)$$

The second is what the experiments themselves forwent: every occupied slot is run at the treatment it was given, so a slot assigned a poor one costs the gap to the optimum for a full cycle of length Δ ,

$$\mathcal{R}_{\text{cum}} = \Delta \sum_{r=1}^R \sum_{u \in \mathcal{U}_r} \left[f_{\theta}(x_{\hat{\theta}}^*) - f_{\theta}(x_u) \right]. \quad (5)$$

This is the usual simple-versus-cumulative pair, and the second term is usually dropped in benchmark settings because a trial there costs nothing. Here an agent that locates the optimum by first running every bad treatment has answered the question and exhausted the facility that paid for it. Neither implies the other, and in our runs they do order agents differently (Appendix I); an agent that runs no experiments forgoes nothing and is judged by (4) alone.

3.3.2 Risk conditioned on the visible history

The common-history evaluator compares design support and recommendation quality against the same information set. Let H_t contain every submitted design, completed result, within-cycle measurement and recommendation visible by simulation time t , and define $R(\theta, x) = f_{\theta}(x_{\hat{\theta}}^*) - f_{\theta}(x)$. The Bayes risk remaining after that history is

$$b(H_t) = \min_{x \in \mathcal{X}} \mathbb{E}[R(\theta, x) \mid H_t]. \quad (6)$$

For the agent’s contemporaneous recommendation $\hat{x}_t$, its posterior risk and excess risk are

$$a(H_t, \hat{x}_t) = \mathbb{E}[R(\theta, \hat{x}_t) \mid H_t], \quad g(H_t, \hat{x}_t) = a(H_t, \hat{x}_t) - b(H_t) \geq 0. \quad (7)$$

The numerical action set always includes $\hat{x}_t$, so the inequality follows from the minimisation rather than from clipping. The conditional value of a newly committed design D is $v(D \mid H) = b(H) - \mathbb{E}[b(H, D, Y_D) \mid H, D]$; it conditions on earlier evidence instead of restarting from the prior each round.

The quantities in (7) replace the Version 1 ratios called design efficiency c and decision efficiency η , which are withdrawn. Their numerators were prior, preposterior design scores while their denominators used realised recommendations, so η was not a fraction of acquired information. A correlation between c and η , movement in opposite directions, or their product does not identify independent design and reader failures. Likewise, identifying θ while sampling far from the optimum need not imply low decision value: the value depends on the posterior action, not the physical distance between sampled treatments and the optimum.

Estimation. Crop physiology and site economics are updated as independent continuous priors from canopy/harvest and cost records respectively; terminal revenue then couples their product posterior. One posterior sample selects the Bayes action and an independent sample evaluates it and the agent recommendation. Plant variation is integrated with common stratified draws, and chamber, loop, batch, measurement and reporting-resolution covariance retain their actual scopes. Finite candidate search provides only a feasible lower bound on each world’s optimum and is labelled as such. We report seeds, effective sample sizes, predictive checks and between-run error, and suppress quantitative decompositions whose calibration error is not below the effect being discussed.

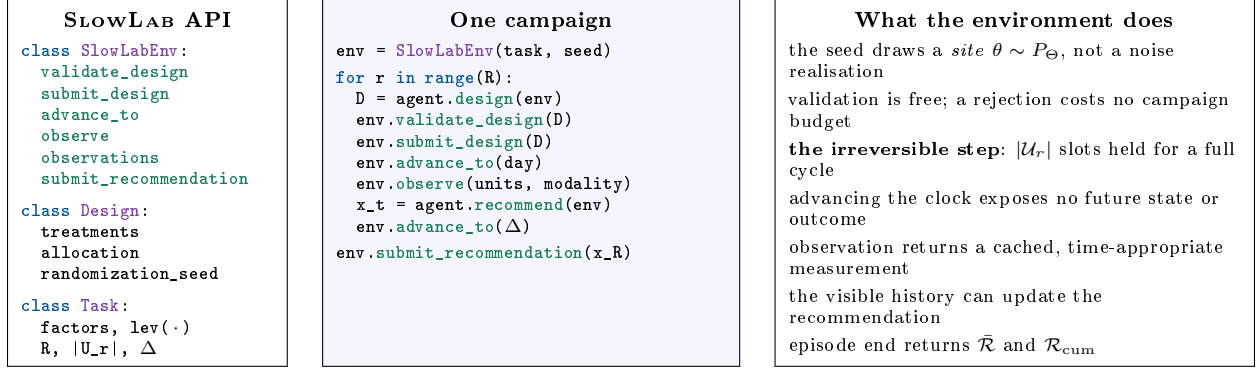


Figure 2: **The Version 2 interaction protocol.** (*left*) The design, clock, measurement and recommendation calls available to an agent. (*centre*) A campaign may contain several inspection times within one committed cycle. (*right*) Committing occupies the units and cannot be undone; advancing exposes no future state; observing reads the cached measurement for the requested day rather than drawing another independent sample. Terminal outcomes remain hidden until Δ .

4 Environment implementation

4.1 The environment

SLOWLAB is a harness: the components below are defined over a domain rather than within one, and §3.1 supplies the first domain that fills them in; Figure 1 on page 2 is the loop this subsection specifies.

4.1.1 Interaction protocol

Figure 2 is the whole interface. `validate_design` is free and unlimited; `submit_design` commits the named slots for one cycle; `advance_to(day)` moves the simulation clock without resampling the crop, and `observe` returns only measurements available on that day. Terminal outcomes are released when the cycle completes. An interim recommendation records what the agent would advise from the history currently visible. Budget is accounted in slot-cycles. The recommendation is elicited, never consumed: it costs no budget, returns nothing, and enters no later design — an agent that revises it every round is not penalised for having been wrong earlier. Design and recommendation are separate submissions, which is what allows an agent to run a good campaign and then answer badly, and §3.3.2 to score the two apart.

4.1.2 Action space

A submission is a `Design`: named treatments, each a full vector of factor values; an allocation mapping treatments to specific units; and a declared randomisation seed. The agent chooses *what* to test, *how many units* each treatment receives, and *which* units those are—replication and allocation are decisions rather than defaults. It may choose an observation day, modality and subset of units from the measurement channels implemented by the task. Version 2 initially exposes cached sensor measurements without a separate measurement charge. It cannot change treatment, terminate, clear or replant a unit mid-cycle; a submitted design runs to completion.

4.1.3 Facility and control levels

A facility is C blocks of L units. Every factor carries a *control level*: a BLOCK-level factor takes one value for a whole block, a UNIT-level factor is set per unit. The level is a physical property of the apparatus, not a modelling choice, and a design assigning two different block-level values inside one block describes an experiment that cannot be performed.

Two consequences follow, and we learned both by implementation. With k block-level factors, the number of distinct block-level *combinations*—not levels per factor—bounds the number of whole-plot treatments by C . And if a combination occupies exactly one block, its effect is confounded with that block’s, and unit-level

replicates cannot be spread. A valid design is therefore a **split-plot**: block-level factors as whole plots replicated across at least two blocks, unit-level factors as subplots. This structure is what makes allocation a decision, and it is absent from any environment in which an experiment is a function call.

4.1.4 Forward model and instances

The environment takes a published forward model f_θ mapping a factor vector to a mean response, and a parameter distribution P_Θ describing how that model’s coefficients vary across sites in the literature. A seed selects an *instance* by drawing $\theta \sim P_\Theta$: a different problem, not the same problem observed again. The response surface and the location of its optimum are therefore fixed by the published model rather than by us.

Within an instance, each unit receives its own draw of the model’s parameters rather than an additive noise term. Unit-to-unit variation *is* parameter variation, and modelling it this way yields heteroscedastic, temporally correlated variability without further assumptions. The nuisance effects of §2.1 are what an allocation must avoid confounding with.

4.1.5 Objective and cost

The objective is a rate: revenue per unit area per unit time, less the full cost of the resources the treatment consumed, $f_\theta(x) = (\text{revenue}(x) - \text{cost}(x))/\Delta$. Pricing is not decoration. Under a pure-yield objective a forward model is typically monotone in its resource inputs, so their optima sit on the boundary and those dimensions are trivial; charging for them is what produces interior optima. Expressing the cost in the objective’s own units is also what makes the *Costly* measure interpretable: a campaign’s expenditure is directly comparable to what it was trying to maximise.

The costs enter at weight one, and this is enforced rather than chosen. An earlier version carried a multiplier λ on the cost term, declared as a task-design parameter. It was not one: it set where every optimum sat, and at $\lambda = 0.25$ it pushed four of six factors onto their rails. The environment now raises if $\lambda \neq 1$ (Appendix J). What TRANSFER moves mid-episode is a price in the cost model, not a weight on it, so the objective stays the profit a grower would actually bank in both halves of that task.

4.1.6 Feasibility checking

`validate_design` checks *mechanical* feasibility only—control levels, unit conflicts, budget, factor ranges—and is free and unlimited; rejections cost no budget but are counted. The design measure of §3.3.2 is computed at scoring time and cannot be called by the agent. Separating the two has a purpose: because feasibility is free to check, a design that passes it yet is scientifically poor is unambiguously a reasoning error rather than an interface error. Anything requiring a statistical assumption—assumed variances, power targets—belongs to the tool layer, an experimental condition to be ablated (Appendix I), not a free service of the environment.

4.1.7 Language-model interface

A language model reaches the same interface through a harness that serialises the round. It receives the facility layout and unit naming, the budget and rounds remaining, each factor’s range and control level, the hierarchy constraint, every time-stamped observation so far with the settings that produced it, the current simulated day, the currently available units, and a JSON schema. During an active cycle it may request a later observation day and measurement channel or finish the cycle; it can update its recommendation from the resulting visible history without changing the committed treatment.

The prompt states constraints and never gives advice: the words replication, randomisation, blocking, split-plot, pure error and confounding do not appear, since those name the capability under test, and a regression test enforces their absence. Failures are counted separately—a reply that will not parse from a design that is mechanically infeasible—both fed back verbatim and retried up to three times, and neither consuming budget, consistent with `validate_design` being free. “Cannot emit valid JSON” and “cannot design an experiment” are therefore different numbers. Exhausting the retries forfeits that round and the campaign continues, so one unparseable reply is not as costly as a failed campaign.

System	noisy outcomes	finite campaign	parallel / batch constraint	time-stamped interim data	separate final recommendation
BoxingGym	yes	yes (10 queries)	not reported	not reported	yes (prediction/model)
Summit	not reported	user-set	not reported	not reported	N/A
Olympus	yes	user-set	not reported	not reported	N/A
Minerva	yes	yes	yes	not reported	N/A
CARE	dataset outcomes	yes (15 total)	N/A (sequential replay)	N/A	best tested condition
SLOWLAB	yes	yes	tiered physical capacity	yes	yes (continuous action)

Table 2: Scope of the evaluated interfaces in the cited work. “N/A” means the published task has a different endpoint; “not reported” avoids inferring a negative from an omitted feature.

4.1.8 Configuration and extension

Configurations are declarative: factors and their control levels, facility shape, rounds, per-round width, variance components, cost weight. Adding one requires no environment code. Agents implement a single `run(env)` method and are registered by name; the reference strategies of Appendix I and the harness above use the same interface, and the harness needs only a callable mapping a message list to a string, so no vendor SDK is required. A new domain supplies f_θ , P_Θ , a cost function in the objective’s units, and a facility whose units are not interchangeable (Appendix A).

5 Related work

Agent benchmarks for experimentation. BoxingGym gives language agents ten sequential experiments in generative probabilistic environments and evaluates experimental design by expected information gain (EIG), followed by prediction or model communication [Gandhi et al., 2025]. Its reported learning curves are especially relevant here: prediction usually improves with more observations, but error increases in the hyperbolic-discounting environment. SLOWLAB asks a different question. The endpoint is an operating recommendation, the design is an allocation to named physical units, and trials consume a fixed campaign budget and opportunity cost. We therefore score decision regret and condition design diagnostics on the same history used for the recommendation, rather than equating parameter information with decision value.

CARE also evaluates an LLM inside a budgeted sequential reaction-optimization loop [Liu et al., 2026]. It has the model write an executable ranking program while a numerical reference policy and intervention gate retain control over experiment selection. Its matched offline replay reveals one condition at a time from an enumerated pool. SLOWLAB instead evaluates direct agent control of feasible batches, including allocation, correlated measurements and a recommendation that need not be one of the tested treatments. These are complementary tests of how an LLM enters an experimental controller.

Executable experiment-optimization benchmarks. Summit provides mechanistic and data-driven reaction benchmarks and common optimizer interfaces [Felton et al., 2021]; Olympus adds noisy emulators trained on experimental chemistry and materials datasets and a broad planner library [Häse et al., 2021]. Minerva goes further toward laboratory execution, supporting noisy multi-objective optimization in large parallel batches and enforcing batch constraints in automated high-throughput chemistry [Sin et al., 2025]. These systems establish that noise, finite data and parallel execution are already central benchmark concerns. SLOWLAB contributes a different resource geometry: controls act at chamber and loop levels, so the agent must assign treatments to units under split-plot constraints, and every tested treatment is charged by its economic shortfall. Table 2 uses “not reported” where the cited evaluation does not specify a property; absence of a report is not evidence that a system cannot represent it.

Budget, delay and experimental structure. Budgeted Bayesian optimization studies how to allocate a total evaluation-cost budget, including the non-myopic case where costs vary and are initially unknown

[Astudillo et al., 2021]. Delayed-feedback BO studies how to choose new evaluations while earlier ones remain pending [Verma et al., 2022]. These methods provide algorithmic baselines and regret analyses; our benchmark fixes a domain-specific facility and exposes the resulting control hierarchy to general-purpose agents. The chamber/loop hierarchy is a split-plot design: whole-plot factors have fewer independent experimental units than subplot factors, so analysis and randomization must respect two error strata [Goos and Jones, 2011]. This statistical structure motivates both the feasibility checker and the block-aware readers.

Decision-theoretic design. Bayesian experimental design commonly scores parameter information [Lindley, 1956, Chaloner and Verdinelli, 1995]; decision-theoretic design instead values how observations change a downstream action [Raiffa and Schlaifer, 1961, Howard, 1966]. Related quantities appear as the knowledge gradient and entropy-search criteria in Bayesian optimization [Frazier et al., 2008, Hennig and Schuler, 2012, Hernández-Lobato et al., 2014]. Our common-history diagnostic applies this decision view to benchmark traces: it separates risk supported by the observed history from excess risk introduced by the submitted recommendation, while reporting the numerical calibration limits of the posterior evaluator.

Scientific agents and self-driving laboratories. Scientific-agent suites often start from existing data, while autonomous chemistry and materials systems connect planners to laboratory tools [Chen et al., 2025, Majumder et al., 2024, Boiko et al., 2023, M. Bran et al., 2024, Szymanski et al., 2023, Burger et al., 2020]. CropGym-style environments expose crop simulators as control problems [Coyle et al., 2025]. SLOWLAB occupies a narrower intersection: it is a simulated benchmark for experimental allocation and final recommendation in one greenhouse domain. We do not infer cross-domain validity from that instantiation.

6 Experiments

6.1 Protocol

We report three evidence layers. First, fixed-history replay crosses scripted design policies with ordinary, block-aware and component-aware GP readers, where a reader is operationally the mapping from a completed visible history to a recommendation. Each reader receives a copy of the same accepted and completed history, so changing the reader cannot change the evidence it reads. Second, the frozen Phase 4 protocol runs closed-loop model agents on reserved sites. Its primary matrix crosses Luna with bare, design-aid, inference-aid and combined interfaces on OPTIMISE: 56 sites, two provider generation seeds and 448 episodes. Prespecified secondary matrices add 96 Sol episodes, 192 Luna episodes on SANITY and TRANSFER, 96 Luna episodes comparing terminal-only and within-cycle access, and 96 prompt-variant episodes, for 928 total. Third, after the primary analysis was frozen, a separately registered robustness extension reruns all four Luna interfaces on 16 fresh sites, two provider generations and noise multipliers 0.5, 1, 2, for 384 further episodes.

A site, rather than an API completion, is the inferential unit. We average the two provider generations within each site, form paired differences, and use a 10,000-resample site bootstrap for percentile 95% intervals. The primary three tool-minus-bare comparisons use Holm family-wise correction. Named secondary families use Benjamini–Hochberg adjustment. Model-format failures, infeasible submissions and early stopping remain outcomes; only transport failures are retried with the same identity. Across the 928-episode frozen run there are no missing or duplicate cells. Appendix F gives hashes, site ranges, API metadata and the distinction between confirmatory, secondary and post-primary analyses.

6.2 Fixed histories separate allocation from evidence use

On held-out Phase 3 sites, the range across scripted design sources is 0.01718 regret, the range across deployable readers is 0.00619, and the design–reader interaction RMS is 0.00209. The relative size is task dependent: the design range is largest on TRANSFER (0.04727), whereas on SANITY the reader range exceeds the design range. This supports treating allocation and history reading as separate capabilities; it does not support a universal claim that one dominates. The block-aware reader also fails to beat the ordinary GP on its independent calibration set (paired normalized-regret improvement -0.0112 , SE 0.0095, 24 pairs), so we retain it as a structural diagnostic rather than designate it a stronger baseline. Figure 3 places the executed design, reader crossing, feedback contrast and evaluator check on one scale-aware mechanism summary.

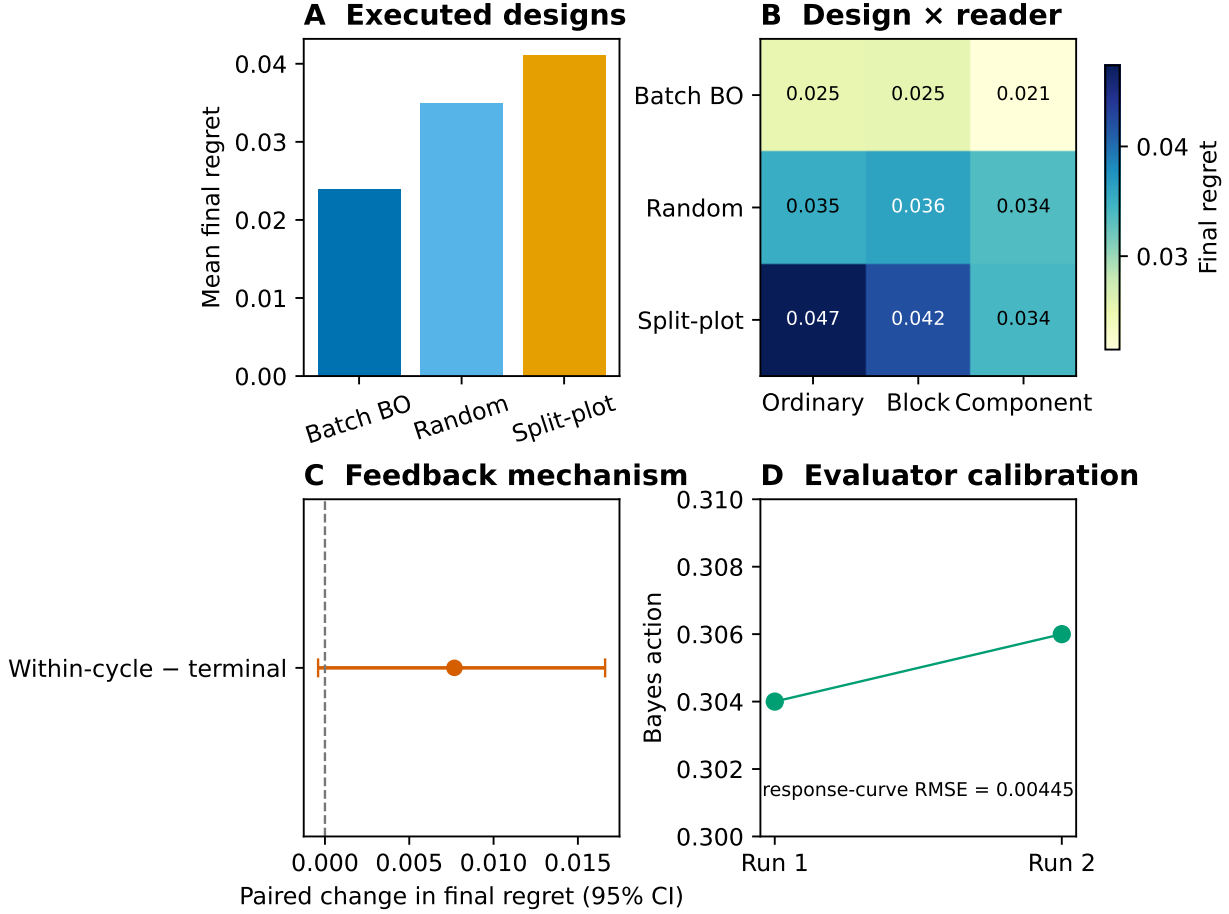


Figure 3: Mechanism checks behind the endpoint comparisons. (A) Mean realized regret of three executed scripted design sources, averaging deployable readers and tasks. (B) The fixed-history design–reader crossing; each cell averages 24 task–site histories, so rows change allocation while columns change only evidence use. (C) The registered paired effect of within-cycle access; positive is worse. (D) Two independent factorized evaluator runs select nearly identical Bayes actions; the annotation reports response-curve disagreement.

6.3 Standard aids do not reliably reduce final regret

Table 3 contains the preregistered primary comparisons. For Luna, the design-aid difference is $+0.00016$ (95% CI $[-0.00299, 0.00321]$), inference aid is -0.00196 ($[-0.00520, 0.00133]$), and both aids are -0.00023 ($[-0.00365, 0.00326]$). None is significant after Holm correction. Sol reaches lower absolute regret than Luna on its independently sampled sites, but its inference aid also does not improve the bare arm: $+0.00053$ ($[-0.00055, 0.00173]$), adjusted $p = 0.381$. Absolute model levels are descriptive because the two models use different reserved sites; the registered inference concerns each paired aid contrast.

The tools were available and often followed. In the Luna primary matrix, the geometric adoption rate is 0.33 for design-only and 0.31 with both aids; inference recommendations are adopted at rates 0.88 and 0.94. The null effects therefore do not arise from complete non-use. They show that these particular tools, delivery format and agents do not reliably convert adoption into lower endpoint regret.

Task and feedback extensions in Table 4 remain uncertain. The inference-aid point estimate is slightly favorable on SANITY and unfavorable on TRANSFER; both intervals cross zero after the prespecified correction. Allowing Luna to inspect time-appropriate within-cycle measurements changes its actions and increases API use, but the final-regret difference against terminal-only access is $+0.00768$ ($[-0.00041, 0.01662]$, adjusted $p = 0.085$). Under the present synchronous action space, committed treatments cannot change and the next batch receives terminal outcomes anyway. Interim observation can therefore update an interim recommen-

Policy	Reader / aid	Sites $\times$ repeats	Mean regret	Δ vs. bare [95% CI]	adjusted p
Random spread	ordinary GP	8×1	0.0072	descriptive	—
Split-plot DOE	ordinary GP	8×1	0.0128	descriptive	—
Batch BO	ordinary GP	8×1	0.0066	descriptive	—
Luna	none	56×2	0.0128	reference	—
Luna	design	56×2	0.0129	+0.0002 [-0.0030, +0.0032]	1.000
Luna	inference	56×2	0.0108	-0.0020 [-0.0052, +0.0013]	0.727
Luna	both	56×2	0.0126	-0.0002 [-0.0037, +0.0033]	1.000
Sol	none	24×2	0.0030	reference	—
Sol	inference	24×2	0.0035	+0.0005 [-0.0006, +0.0017]	0.381

Table 3: Final simple regret on OPTIMISE (lower is better). Scripted rows are fixed-history reader replays on eight Phase 3 sites and are descriptive; they are included as reference points, not tested against the independently sampled model runs. Phase 4 model contrasts average two provider generations within each site before inference. Luna tool contrasts use Holm correction across the three preregistered comparisons; the Sol replication uses the prespecified secondary-family adjustment.

Contrast	sites	Δ regret	95% CI	adjusted p
Luna, T1: inference — bare	24	-0.0008	[-0.0053, +0.0037]	0.722 (BH)
Luna, T4: inference — bare	24	+0.0114	[-0.0017, +0.0287]	0.313 (BH)
Luna, within-cycle — terminal-only	24	+0.0077	[-0.0004, +0.0166]	0.085 (BH)
Luna, $0.5\times$ noise: design — bare	16	+0.0113	[+0.0053, +0.0184]	0.010 (BH)
Luna, $2\times$ noise: design — bare	16	+0.0048	[-0.0047, +0.0131]	0.605 (Holm)
Luna, $2\times$ noise: inference — bare	16	-0.0067	[-0.0155, +0.0012]	0.374 (Holm)
Luna, $2\times$ noise: both — bare	16	-0.0040	[-0.0116, +0.0032]	0.605 (Holm)

Table 4: Held-out secondary and robustness contrasts (positive values are worse). The half-noise design-aid contrast is the only listed interval that excludes zero.

dation, but it has no guaranteed path to improve the next design. Testing value from inspection requires a consequential action such as staggered release, early termination or treatment modification.

6.4 Robustness changes the conclusion only at lower noise

The fresh adaptive extension asks whether observation noise changes later designs and recommendations, which fixed-policy rescoring cannot test. At twice the registered noise, all three tool-minus-bare intervals cross zero (Table 4), matching the primary conclusion. The prespecified secondary family gives a different result at half noise: design aid increases regret by 0.01133 (95% CI [0.00530, 0.01842], adjusted $p = 0.0096$). Inference-only and combined aids at half noise, all one-noise contrasts, and the registered $2\times$ versus $1\times$ interactions are not significant. We therefore report a bounded conclusion: the tested tools show no reliable improvement, and one lower-noise condition shows harm; the evidence does not establish a universal direction across models, domains or tool implementations.

6.5 Transfer is evaluated after the price shock

Table 5 reports the endpoint the task is designed to test. Both arms receive the energy-price shock only after completing their experimental campaign and must recommend again without new trials. The inference-aid arm has higher post-shock regret by 0.0452 on average, with a descriptive paired site-bootstrap interval [0.0200, 0.0781]. This contrast was not a registered inferential family, so it is evidence for a targeted replication rather than a confirmatory tool-effect claim.

6.6 Evaluator scope

The common-history evaluator is calibrated on SANITY: independent 32×32 crop–economics product posteriors agree to response-curve RMSE 0.00445, and action selection and evaluation use disjoint posterior draws.

Condition	sites $\times$ repeats	pre-shock	post-shock
Luna bare	24×2	0.0602 ± 0.0054	0.0681 ± 0.0095
Luna inference	24×2	0.0716 ± 0.0074	0.1133 ± 0.0140
Inference $-$ bare (post-shock)	24 sites	$+0.0452$ [$+0.0200$, $+0.0781$]	

Table 5: TRANSFER regret before and after the held-out $2.2\times$ energy-price shock (mean $\pm$ site-level SE). The post-shock paired interval is descriptive because this contrast was not a preregistered inferential family.

That check does not establish the same error in the higher-dimensional tasks. For this reason, the primary model and tool claims above use realized final regret, which is computed directly from the hidden site, while $b(H_t)$ and $g(H_t, \hat{x}_t)$ are used as mechanism diagnostics only where their between-run error is smaller than the effect being interpreted.

7 Limitations

The empirical scope is one simulated greenhouse-tomato domain. The mechanistic crop model improves traceability, but the economic coefficients, site distribution, factor ranges, facility layout and observation channels are benchmark choices. Results do not establish the same tool effects in chemistry, materials, biology, field agriculture or a physical greenhouse.

The model study covers Luna and a smaller Sol replication through one provider interface, one base prompt family and passive tool delivery. Provider outputs varied at temperature zero, which is why the protocol treats generation seeds as within-site replicates. A null effect for these aids does not show that statistical tools cannot help an agent; it bounds the tested implementation and delivery mechanism.

Within-cycle observations are consequential only through the recommendation during the current synchronous protocol. Units cannot be released, treatments cannot be modified, and the next batch begins after terminal outcomes are available. The experiment therefore tests whether inspection changes reasoning and endpoint recommendations, not the value of early stopping, asynchronous replacement or adaptive treatment. Those actions require explicit cleanup time, residual value and staggered-capacity semantics.

The common-history evaluator has a strong one-factor calibration but incomplete higher-dimensional calibration. Its finite candidate search also supplies a feasible lower bound on the world optimum. We consequently base primary comparisons on realized regret and restrict posterior-risk decompositions to effects larger than measured evaluator error.

Finally, the Phase 3 scripted reference rows and Phase 4 LLM rows use independent held-out site ranges. They provide common-scale descriptive anchors, but they are not paired model-versus-baseline tests. Direct superiority claims require a jointly randomized matrix on the same sites.

8 Conclusion

SLOWLAB evaluates an agent as the operator of a resource-limited experimental campaign: it must allocate treatments to a tiered facility, interpret time-stamped evidence and issue a final recommendation while poor experiments incur economic loss. Version 2 makes within-cycle observation explicit and evaluates designs and recommendations against the same visible history.

The experiments give a narrower and better-supported result than the Version 1 mechanism story. Fixed-history replay shows that allocation and reading are empirically separable but task dependent. In 928 frozen-protocol episodes, standard design and inference aids do not reliably reduce final regret for the tested Luna and Sol conditions. A fresh 384-episode noise extension reproduces that conclusion at twice the observation noise and finds a harmful design-aid effect at half noise. Within-cycle access is used by the agent but does not significantly improve the final recommendation under a synchronous action space.

These findings motivate benchmarks that expose the resource and timing choices through which better reasoning can affect an experiment. The next decisive extensions are matched scripted-and-model comparisons on the same sites, a calibrated higher-dimensional history evaluator, and asynchronous actions whose use of interim evidence can change physical capacity rather than only an interim recommendation.

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A Admitting a new domain

SLOWLAB is a harness for a class of problems rather than for one domain. A domain is admissible when it supplies four things.

The conditions must hold jointly and by nature, not because the benchmark author chose hyperparameters that way (§3.1).

A published forward model must exist, so that the response surface, its noise structure and the location of its optimum are fixed outside the benchmark author’s control.

A cost model in the objective’s own units must be available, so that exploration is priced rather than constrained from outside.

Experimental units must not be interchangeable. Some structure—blocks, batches, chambers, cohorts—must constrain which treatments can be assigned where, because that is what makes allocation a decision rather than bookkeeping, and what makes replication compete with coverage.

Greenhouse horticulture, the instantiation built here, satisfies all four (§3.1). Microbial evolution experiments, bioprocess scale-up, materials ageing and dose-finding trials meet them with published forward models already in hand. We do not claim results transfer across domains—that is the question a multi-domain version would exist to answer—only that the harness, the measures and the reference points are defined at the level of the class.

Four extensions that need no environment code. The crop model already contains the mechanism each of them tests, so each is a task specification. *Equipment faults* give ground-truth attribution: a heater fails in one chamber for k days, the agent sees an unexpectedly low yield and an anomalous log, and must name the cause, which is exact because we injected it. *Fruit abortion* above T_{CRIT} already makes irreversibility graded rather than binary, from recoverable through partially recoverable to a terminated unit. *Between-season parameter drift* within the published between-site spread makes conclusions expire on a schedule the data rather than we set. And because the oracle is computable in a tenth of a second, a *target profit rate* sorts instances into comfortably feasible, feasible only in a narrow region, marginally infeasible and clearly infeasible, which tests whether an agent can recognise that it has been asked for something impossible. None of these can be built on a surrogate without injecting the abstraction by hand.

B A worked round

Figure 4 runs one round of OPTIMISE through in full: one site, one twelve-unit budget, two designs that both pass the feasibility check and differ only in how the units are allocated. It is the smallest case in which the control hierarchy of §4.1.3 decides the answer, and the reason the environment scores allocation rather than treatment choice alone.

C Task specifications

Factor	Range	Control level	Role
day temperature	18–32 °C	chamber	heating/ventilation setpoint
night temperature	12–22 °C	chamber	setpoint
CO ₂	350–1200 ppm	chamber	enrichment
PAR	18–32 mol m ^{−2} d ^{−1}	chamber	natural + supplemental
plant density	2.2–4.5 m ^{−2}	loop	planting decision
LAI _{max}	2.0–5.0	loop	pruning target

Table 6: The six management factors. Control level is a physical fact—lamps and heating serve a whole chamber, whereas density and the pruning target are set per hydroponic loop—and it determines each factor’s parallelism ceiling.

The noise column of Table 1 is

$$\sigma_{\text{tot}}/\sqrt{B} = \sqrt{\text{sd}^2\left(\frac{\tau_{\text{loop}}^2}{B} + \frac{\tau_{\text{chamber}}^2}{CR} + \frac{\tau_{\text{batch}}^2}{R}\right)},$$

the standard error of the mean of $B = R \times |\mathcal{U}_r|$ observations spread over C chambers and R rounds. Only the first term falls with B : a chamber effect is shared by every unit in that chamber and a batch effect by every unit in that round, so past a point buying more units stops buying precision. That is the width constraint of §2.1 in arithmetic.

Two quantities a grower does control are declared by the task rather than exposed as factors, and the environment refuses to treat either as one. Cycle length is fixed at 210 days: it is monotone over any plausible range, because fruit set does not begin until roughly 30–45 days and that establishment period is pure cost, so exposing it would add a dimension with the same answer at every site. Leaf removal is fixed at 2.25 g node^{−1} because the reduced model gives defoliation no photosynthetic penalty, so its optimum is always at the upper rail. Both are asserted rather than silently overridden, so anyone who adds them back gets an error instead of a sweep that quietly holds them constant.

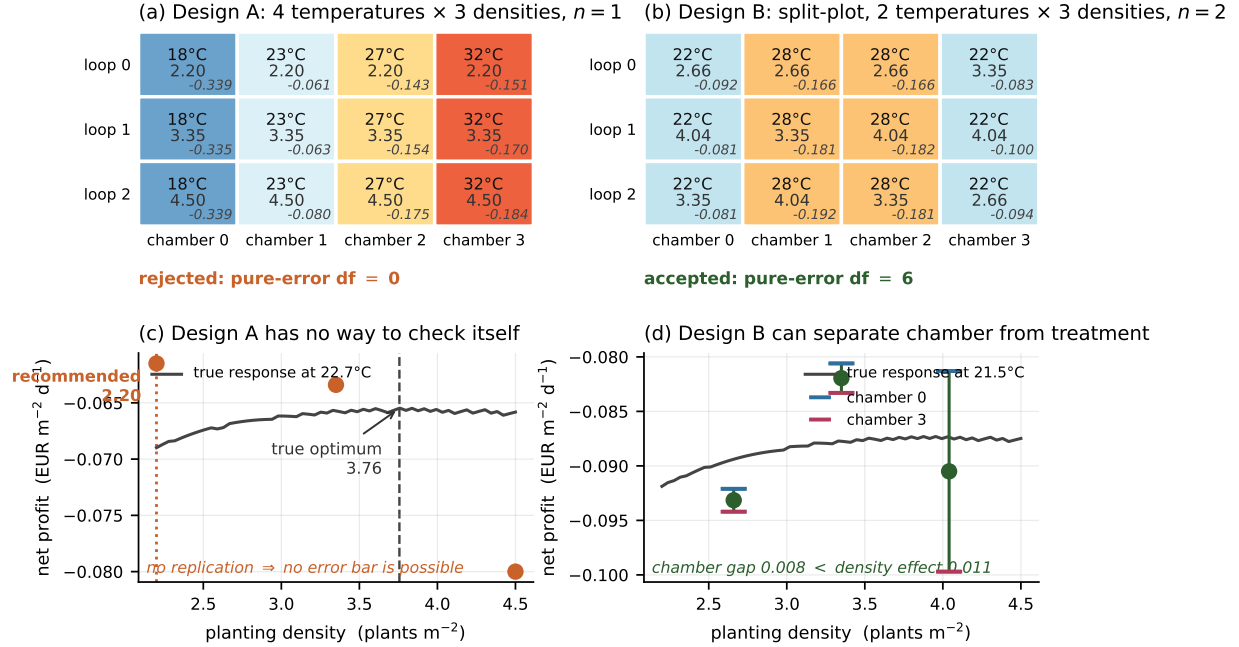


Figure 4: One round of the running scenario: same site (`seed=0`), same twelve-unit budget, two designs. All values are produced by the environment; the script is in the repository. (a) The intuitive design fills the grid—four temperatures, one per chamber, crossed with three densities. It passes `validate_design`, but every treatment is observed once, so pure-error df is zero and each temperature is confounded with one chamber. (b) The split-plot: two temperatures, each replicated across two non-adjacent chambers, densities randomised within chamber. Our first attempt used chambers 0–1 and 2–3 and was rejected by the randomisation check—temperature then correlates perfectly with chamber index. (c) Design A’s best observation sits at 3.35 plants m⁻² where the true optimum at that temperature is 3.76; the miss is small, and the point is that with one observation per treatment nothing in the data says whether it is small. (d) Design B can answer the question it was given: the chamber 0 against chamber 3 gap at identical settings is 0.004 against a density effect of 0.009, so the effect is resolvable above the block noise — which is a statement design A cannot make either way. The figure is regenerated from the frozen environment; an earlier version hard-coded the annotations and printed the comparison in (d) with the inequality facing the other way.

D The naive corner

Setting every factor to its upper bound yields *negative* profit at every site, worse than the site optimum by a factor of 4 to 52; setting every factor to its midpoint is also negative at most sites, worse by a factor of 1.3 to 21. Maximising everything fails because day temperature at its ceiling aborts fruit set (eq. 8) while supplemental lighting continues to draw power. We use this rather than “every factor has an interior optimum” as the criterion for a non-trivial search problem.

E Reproducibility

All results use seeds 0.. $N-1$; `SlowLabEnv(task, seed)` is deterministic, and the seed selects a sampled site rather than only a noise realisation. Every table and figure in this paper is emitted by a script in `scripts/` from the JSON under `results/`, and none is hand-edited. The files carry the environment version in their names, so a table built from a previous environment cannot be picked up silently. Code, task definitions and generation scripts are included in the anonymous supplement. The public Version 1 release contains the archived 800 language-model transcripts; it does not contain the cached atom sets. Version 2 transcripts and compact analysis products are packaged with the submission artifact, with raw prompts separated from the private review notes used during development.

F Phase 4 protocol and execution audit

The confirmatory protocol identifier is `slowlab-v2-phase4-confirmatory-2026-09-13-a2`. It freezes model identifiers, temperature, token limits, prompt variants, site and provider-generation seeds, primary and secondary matrices, missing-cell handling, multiplicity correction and the no-efficacy- stopping rule. Realized prompt hashes and provider metadata are recorded per episode. The primary Luna matrix contains 448/448 expected rows; all frozen matrices together contain 928/928 rows, with no duplicate identities. The post-primary noise extension is `slowlab-v2-phase4-noise-robustness-2026-09-19-a2` and contains 384/384 rows. The compact summary used by the manuscript is `results/phase5_paper_summary_env2.0.0.json`; the LaTeX tables are generated by `scripts/build_phase5_paper_artifacts.py`.

G The $\mathcal{R}^*$ estimator

Version 2 status. This section documents the withdrawn Version 1 estimator for auditability. The main text no longer reports c or η . Version 2 uses continuous crop and economic posteriors conditioned on the common visible history, followed by independent action-selection and evaluation samples; implementation and calibration details are in the accompanying Version 2 protocol.

Observation likelihood. Because the per-unit draw ϕ_u is pushed through the forward model, $p(y \mid \theta, x)$ has no pointwise density. We moment-match the per-unit response to a Gaussian with variance $s^2(x)$, which makes the covariance of a round’s $|\mathcal{U}_r|$ observations

$$\Sigma(D_r) = \text{diag}(s^2(x_u) + \tau_\gamma^2) + \tau_\beta^2 Z Z^\top + \tau_\epsilon^2 \mathbf{1}\mathbf{1}^\top,$$

with $\tau_\beta, \tau_\gamma, \tau_\epsilon$ the nuisance scales of §2.1 and $Z \in \{0, 1\}^{|\mathcal{U}_r| \times C}$ the block-membership matrix. The allocation enters through Z , so where a design places its treatments changes (6), not only which treatments it chooses.

Three approximations. (a) The Gaussian moment-match above. (b) $s^2(\cdot)$ is computed once at the nominal parameter vector rather than per atom. (c) The prior is the M -atom mixture rather than the continuous P_Θ . We had argued that (c) was benign, on the grounds that the dimension which matters is that of x_θ^* rather than of θ , and that it is small — over 150 instances of OPTIMISE the true optima occupy ten grid cells at $H[p(x_\theta^*)] = 2.05$ nats. That argument is wrong, and the next two paragraphs are what it cost. The support of x_θ^* being small does not make the *likelihood* over atoms well behaved, and it is the likelihood that fails. We do not have separate validation for approximations (a) and (b); this is one reason Version 2 does not use the withdrawn estimator for its primary comparisons.

The in-sample estimator and why we abandoned it. Our first estimator drew one atom set and used it for everything: the posterior weights $w_m(y)$, the minimisation over candidate recommendations, and the evaluation of the chosen recommendation. The failure is mechanical. $\Sigma(D_r)$ above contains observation noise and block structure and nothing for the coarseness of the grid, so with 12 to 24 observations the likelihood discriminates between atoms far more sharply than the grid spacing warrants. Measured over 160 atoms on OPTIMISE, the mean effective sample size $1/\sum_m w_m^2$ is 1.1, and the candidate that minimises $w^\top D$ is some atom’s own optimum in 98% of draws. The estimator was therefore answering “can the data identify which of these 160 sites generated it” and reporting the answer as though it were “how much decision risk does this design remove”. Held out on atoms that did not generate the data, the same designs score $c = -64\%$ where the in-sample number reads $+97\%$: conditioning on the design leaves the reasoner worse than the prior-optimal constant, because the posterior concentrates on a wrong atom and the recommendation follows it there.

The estimator we report. Three changes, all in `achievable_regret_cv`. The likelihood is raised to the power $1/T$, equivalently $\Sigma \rightarrow T\Sigma$, which is the minimal way to state that the true site lies between atoms. T is selected by leave-one-out over the inference atoms: for each atom m , draw y from m , form the posterior from the others, choose a recommendation, and score it at m ; take the T minimising the average.

The evaluation atoms are never touched, so selecting T cannot import the bias it removes, and the leave-one-out curve tracks the held-out curve closely enough to be used as its proxy. Finally the recommendation is scored on the disjoint half. The result is c between 1% and 20% where the in-sample estimator read 35% to 78%, and the orderings differ: on OPTIMISE the in-sample estimator ranked random search first among the reference strategies and the repaired one ranks it last.

We tried the obvious cheaper fix first. Fitting the surrogate hyperparameters is not it and neither is raising M alone; tempering is what moves the number, and the temperature the calibration selects is 3 to 100, meaning the effective observation covariance is one to two orders of magnitude larger than the noise model alone implies. That is the size of the discretisation error, stated in the only units available.

Convergence in the atom count. Table 7 runs seven designs at $M = 80$ and $M = 320$ on the two tasks we could afford at both counts, six runs each. The designs span the range of c rather than compete: space-filling, a two-level factorial, four, two and one distinct treatments, and the two degenerate cases of every unit at a corner and every unit at the centre.

The estimator has not converged and the direction of its error is now the opposite of the in-sample one. Quadrupling M leaves the degenerate designs where they are and lifts the good ones: on SCREEN the space-filling design goes from 0.8% to 16.5%. The reason is the split. At $M = 80$ only 40 atoms carry the posterior, which is too coarse a stand-in for P_Θ for any design to separate, so the measure reports that no design helps. Every level we report is therefore a lower bound on what a converged estimator would give, which is the conservative direction for claims of the form “no agent reaches the ceiling” and the awkward one for any claim that a particular agent’s design was poor.

Rank correlation between the two counts is 0.82 on OPTIMISE and 0.64 on SCREEN. That is lower than the in-sample estimator reported and it is the honest number: the in-sample estimator’s near-perfect rank stability was itself an artefact, since a posterior that always collapses onto one atom is a deterministic function of design geometry and therefore stable in M for the wrong reason. A single estimate carries a standard deviation of 2 to 4 points. Every c in this paper is a mean over dozens of designs, where that falls below one point, and the comparisons we draw conclusions from are paired at fixed M .

Design	OPTIMISE		SCREEN	
	c at $M=80$	$M=320$	c at $M=80$	$M=320$
space-filling	21.7 ± 1.3	29.6 ± 2.0	0.8 ± 0.9	16.5 ± 0.8
four treatments	2.2 ± 1.6	16.0 ± 1.4	0.5 ± 1.1	7.8 ± 0.8
two treatments	4.4 ± 1.5	12.2 ± 1.5	12.9 ± 0.9	7.8 ± 1.1
two-level factorial	-2.0 ± 1.8	-0.5 ± 2.1	-1.7 ± 0.4	3.4 ± 1.1
corner only	-1.9 ± 1.5	-1.7 ± 2.1	0.7 ± 0.8	1.5 ± 0.8
centre only	-0.4 ± 1.6	-1.9 ± 1.7	-0.4 ± 1.3	-3.1 ± 0.5
one point	-2.0 ± 1.6	-2.3 ± 2.1	-0.7 ± 1.1	0.6 ± 0.6
$\mathcal{R}^*(\emptyset)$: OPTIMISE $0.00546 \rightarrow 0.00610$; SCREEN $0.00500 \rightarrow 0.00443$				
rank correlation of c between the two atom counts: 0.82 and 0.64				

Table 7: Design efficiency at two atom counts under the cross-validated estimator, seven designs chosen to span the range rather than to compete. Each entry is the mean of six independent runs with its standard error; a single run carries a standard deviation of 2 to 4 points, which is why the levels reported elsewhere in this paper are always means over dozens of designs. Quadrupling the atom count leaves the two degenerate designs where they were and roughly doubles the best one on OPTIMISE; on SCREEN it moves the space-filling design from nothing to the top. The estimator is not converged, and the direction of its error at low M is to under-credit good designs, because half of 80 atoms is too coarse a stand-in for P_Θ for any design to separate.

The bound does not survive estimation. For the exact quantities $c \in [0, 1]$ is a theorem: numerator and denominator start from the same beliefs, so $\mathbb{E}[\min] \leq \min[\mathbb{E}]$ gives $\mathcal{R}^*(D_r) \leq \mathcal{R}^*(\emptyset)$, and $\mathcal{R}^* \geq 0$ closes the other end. The estimator above is a tempered rule graded out of sample, about which the optimality

theorem says nothing, and the difference is visible: across the 320 archived Version 1 reference rounds, 69 carry a negative c . Averaged over the dozens of rounds that make up one cell every value we report is inside $[0, 1]$ — c from 1% to 20%, η from 4% to 56% — but by margin rather than by proof. For η there is not even an exact bound to appeal to, since the agent’s $\hat{x}_\tau$ has seen every round while the numerator scores each round separately against the prior.

Trajectory and endpoint. The interim recommendation elicited after each round uses the same rule the agent applies at the end, so the last point of a trajectory equals the regret of the submitted recommendation. This is asserted by a regression test: an earlier version used the best observed point for the trajectory and a posterior maximum for the submission, which differed by 47% on OPTIMISE.

H The horticultural instantiation

Forward model. f_θ is the reduced state-variable TOMGRO model [Jones et al., 1999]: five coupled state variables—mainstem nodes, leaf area index, above-ground, fruit and mature-fruit dry weight—implemented from its equations (1)–(11) with the published least-squares estimates. We deliberately do not sample ground truth from a Gaussian-process prior: such a sample has no physical content, its length scale would be our choice, and it makes the modelling assumption of GP-based agents exactly correct. Fitting our Bayesian-optimisation reference’s own surrogate to the mechanistic response gives mean absolute error $1.1\times$ the global average in the high-temperature decline and $2.2\times$ near the kink where supplemental lighting begins to incur cost—real but mild, and the surrogate still locates the optimum. What the mechanistic surface supplies that a stationary GP cannot is structure: fruit abortion above T_{CRIT} (eq. 8), the source of irreversibility, and a source-sink limit taking the *minimum* of two growth expressions (eqs. 9–10). P_Θ follows the same paper’s finding that vegetative parameters are stable across sites while fruit parameters vary: each seed draws fruit parameters across the published between-site spread (N_{FF} varies by $2.2\times$ across their three sites). Five of the six factors have optima that move across sites; the sixth, supplemental light, sits at the lower rail in every instance because at 2024 European electricity prices it never repays its energy.

Facility. Blocks are climate chambers and units are hydroponic loops. Lamps, heating and CO_2 enrichment are set per chamber; planting density, pruning target and the pruning target is set per loop. Cycle length is fixed at 210 days for every unit and is not a decision (§3.2).

Costs. Revenue is fruit fresh weight at wholesale price; costs are supplemental lighting, heating and CO_2 enrichment, computed with standard greenhouse engineering relations. Coefficients come from public sources rather than from us, because the *Costly* measure is denominated in euros and an uncited coefficient makes its absolute value uninterpretable: crop price 1.50 EUR/kg fresh weight, between Dutch (1.57) and Spanish (1.35) vine tomato in March 2024 [HortiDaily, 2024]; electricity 0.189 and natural gas 0.0605 EUR/kWh, the EU non-household averages for 2024 [Eurostat, 2025]; luminaire efficacy $2.5\mu\text{mol J}^{-1}$, the level over half of listed horticultural fixtures reach [DesignLights Consortium, 2022]; a cover heat-transfer coefficient of $4.0\text{ W m}^{-2}\text{K}^{-1}$, between single glass (≈ 4.5) and double polyethylene (2.9–3.4) [Kittas and Baille, 1998]. The CO_2 leakage coefficient yields 14.6 kg m^{-2} per season, inside the 12–24 range that must be delivered for tomato [AHDB, 2020].

Caveats. The figures are representative European values, not measurements from one facility. Outdoor temperatures correspond to the Mediterranean-subtropical winters of TOMGRO’s calibration sites rather than to northern Europe, where greenhouses consume $47\text{--}69\text{ m}^3$ of gas per m^2 per year ($\approx 460\text{--}670\text{ kWh}$) against about 245 here—a difference of climate rather than of coefficients. And one consequence was not our choice: at these prices supplemental lighting is never economical in this climate, so the optimum for PAR sits at its lower bound for every site.

I Scripted reference policies

The Version 2 reference set contains random spread, split-plot DOE and constraint-aware batch Bayesian optimization. Random spread maximizes geometric coverage subject to the same facility validator as model

agents. Split-plot DOE balances whole-plot combinations across chambers and randomizes loop-level settings within each chamber. Batch BO fits an ordinary GP to completed outcomes and proposes a feasible batch by sequential fantasization. Readers are evaluated separately by copying each accepted, completed history; ordinary GP, block-aware GP and component GP then recommend without changing any design or observation. Table 3 reports the ordinary-GP fixed-history reference points on OPTIMISE. Their sites are independent of the Phase 4 model sites, so they are descriptive anchors rather than paired superiority tests.

J Version 1 defect register and remaining scope limits

Version 1 is frozen as an audit artifact. Version 2 fixes the claim-changing defects below and is the environment used for the new experiments; the remaining entries bound what that environment represents rather than certify the simulator as a physical greenhouse model.

The stopping rule. *A defect is fixed if it changes a claim the paper makes, and recorded if it does not.* We adopted this after the alternative became visible: while $\mathcal{R}^*(\emptyset)$ was being used as an admission test for tasks, every correction to the crop or cost model moved the ceiling, which moved the task set, which motivated the next correction. That loop does not terminate, because a simulator can always be made more nearly right. Reporting ranks rather than levels (§3.3.2) removes the dependence and lets the rule bite.

The rule caught four defects and let three stand. Four defects changed claims and were fixed: a weight λ on the cost term, nominally a task-design parameter, which set where every optimum sat and pushed four of six factors onto their rails at $\lambda = 0.25$; a photosynthesis routine that read global nominal parameters instead of the site’s, so that every site had an identical light and CO_2 response and a stated limitation about sampling those parameters was a no-op; a heating model that applied $AU\Delta T$, the peak-sizing formula for the coldest night, to all 24 hours of 210 days, overstating a term worth 48% of variable cost; and an objective named net profit that omitted harvest labour, grading loss and all fixed costs. One defect was recorded instead: the block effects below. The Version 1 ratio they entered has been withdrawn; Version 2 retains their physical scopes in its observation model.

Noise model. The block scales τ_{chamber} , τ_{loop} and τ_{batch} multiply the standard deviation of the objective over uniformly random points in the whole factor space, a quantity dominated by corners no grower would operate in. Improving the ventilation physics left the optimum unmoved to three figures and inflated that standard deviation by 71%, and with it the chamber effect, from 4.4% to 7.9% of revenue at the optimum. A chamber’s climate uniformity has nothing to do with how bad the worst corner of the design space is; the effect should be a temperature offset pushed through the forward model, with a magnitude taken from measured greenhouse uniformity. Separately, `task.noise_sd` is a declared value 1.3 to 1.9 times the per-unit standard deviation the simulator actually produces; the simulator is right and the declared value is unused except in reporting.

Crop and cost model. The profit-optimal yield averages $28 \text{ kg m}^{-2} \text{ yr}^{-1}$ over sites, range 21 to 37, against a commercial 50–70, while the model’s *maximum attainable* yield is 59, inside that range: the model can grow a commercial crop and does not think it pays. We have not established why. Supplemental light is rail-bound at every site, so the entire lighting sub-model has zero elasticity on the objective. This is also what lowers the SCREEN ceiling, and it separates two questions a sensitivity analysis conflates: over 24 sites, setting supplemental light mid-range rather than at its optimum costs 0.080, the largest of the six factors, while setting it at the across-site mean optimum — all an experiment could still recover — costs 0.000, because its optimum does not move. The factors worth an experiment are day temperature and CO_2 , at 0.0085 and 0.0060, together 85% of what instance-tailoring can recover there (`scripts/factor_value.py`). Relatedly, `par_ambient` is fixed across sites although natural daily light integral is the largest real difference between a Dutch and an Almerian greenhouse. A daily-timestep model cannot represent a time-varying CO_2 enrichment strategy, which is what growers actually run. The ventilation capacity is clipped rather than allowed to fail, so overheating above that cap is not modelled.

Provenance of coefficients. Electricity and gas prices, the site-level spreads, the LED efficacy range and the cover heat-transfer range are cited. Four numbers are not: the tomato price, whose source we can no longer verify and which carries the largest elasticity on the objective; the fruit dry-matter fraction, which has no citation at all and carries the second largest; the transplant cost, quoted in dollars and used as euros; and two labour coefficients whose derivations contain wage and task-share figures that are ours. One property limits the damage: the tomato price and the dry-matter fraction scale revenue linearly and $\mathcal{R}^*$ scales with them, so they cancel from every ratio we report. $\mathcal{R}^*(\emptyset)$ is 3.1% of revenue at the optimum on both SANITY and OPTIMISE, and adding harvest labour and grading loss left that ratio where it was. These coefficients set the euro scale and no conclusion.

Estimator and action space. The estimator’s own defects, and the one we found and repaired, are in Appendix G; they are the largest single item in this register. Separately, replication does not pay at these budgets, so an agent is never rewarded for buying precision — an arithmetic consequence of bounded parallelism rather than a bug, but it narrows what the benchmark can distinguish.

A stronger form of irreversibility is implemented and not enabled. All four tasks set the recovery length to zero. The code path exists, but the Version 2 experiment matrix does not evaluate recovery delays, so no claim depends on that stronger mechanism.

Two actions a real grower has and this agent does not. The action space stops at a design and a recommendation, and two of the omissions matter more than the rest.

Early termination. A greenhouse is watched continuously—climate at minute resolution, crop registration weekly, harvest accumulating from about week ten—so a grower who has planted a bad treatment usually knows by week six and pulls it. We have no such action: a committed unit runs its full 210 days. The omission is what makes §2’s two readings of delay, information and action, coincide here; restoring it would separate them, and it is the cleanest available trade-off between slowness and irreversibility, since salvaging costs part of a season rather than all of it.

Staggered planting. Rounds are synchronous: every unit in a round starts and ends together. Real trial programmes run a pipeline, starting a new batch every few weeks. Our budgets leave half the facility idle on two of the four tasks—8 of 16 unit-slots on SANITY and 12 of 24 on SCREEN—for the full 420 days, which no grower would accept. Staggering at a 105-day offset would run four batches instead of two in the same wall-clock. It also poses the harder question: with only partial information from batch one, what do you plant in batch two? That is where slow feedback actually bites, and it is the first thing we would add.

Version 2 now lets an agent choose observation time, modality and unit subset. It does not yet price those measurements or let the agent change how many plants constitute a physical unit.